CYBR 345 H14-HW

Instructions:

- Do not delete, edit, or change any text or questions in this document.
- Type or write your answers directly below each question.
- Show all work and clearly explain your reasoning using complete sentences and sufficient detail.
- For each part of the homework, answer all subparts carefully and support your responses with observations from the activity and concepts from class. A well-developed answer will usually be several sentences long.
- When inspecting certificates, VPN explanations, or Wi-Fi connection settings, include screenshots (snips) as evidence of your work. Make sure the relevant details are clearly visible.
- If submitting handwritten work, ensure images are clear, properly oriented, and easy to read. 2
- Submit your work only through Blackboard; do not email it.
- Assignments that do not show required work, include screenshots where required, or follow instructions will receive a grade penalty or a grade of 0.

Objectives:

- Explain how cryptography is used in real-world systems such as TLS, VPNs, Wi-Fi, and Kerberos
- Analyze how secure connections are established and maintained
- Identify the roles of encryption, authentication, and integrity in different systems
- Interpret real-world security artifacts such as certificates and Wi-Fi security settings
- Evaluate security risks and protections in practical scenarios
-

Part 1: TLS & Certificates (25 pts)

Visit a secure website and inspect its certificate using the browser lock icon.

Chosen Website: Official New York Islanders Website | New York Islanders

(a) Identify who issued the certificate and who it is issued to, and explain what information the certificate contains and why it is important.

The certicicate for www.nhl.com/islanders is issued to www.nhl.com and issued by WE1, a part of Google Trust Services. It includes the domain name, issuing Certificate Authority, public key footprint, certificate footprint, signature information, and the validity period (04-15-2026 to 07-14-2026). This information is important because it allows the browser to verify that the website is legitimate and the public key belongs to the NHL website. The certificates ensure identity verification and allow for encrypted communication during the TLS handshake.

Issued To

Common Name (CN)	www.nhl.com
Organization (O)	<Not Part Of Certificate>
Organizational Unit (OU)	<Not Part Of Certificate>

Issued By

Common Name (CN)	WE1
Organization (O)	Google Trust Services
Organizational Unit (OU)	<Not Part Of Certificate>

Validity Period

Issued On	Wednesday, April 15, 2026 at 1:48:05 AM
Expires On	Tuesday, July 14, 2026 at 2:48:03 AM

SHA-256 Fingerprints

Certificate	bd2f321badfeaca4318124a3b250b5d9a036414972fde4cff98ef2616548bb9f
Public Key	4b85db1fec586687dcc4829dc22c18a274357322fec4443399c7594f636be4d5

(b) Locate the public key and explain its purpose, including why the browser trusts this certificate.

The public key is availitble under the certificates section of the inspector and in this case uses SHA-256. The public key allows the browser to encrypt the key-exchange information so that only the NHL's server can decrypt it using the private key, ensuring that attackers can't view or modify the handshake. The browser trusts the certificate because it's signed by Google Trust Services, which is the CA that is already within the browsers trusted root store. The browser verifies the CA signature and confirms that the domain name matches, which validates the certificate and establishes a secure connection.

bb9f

Public Key	4b85db1fec586687dcc4829dc22c18a274357322fec4443399c7594f636be4d5

(c) Explain when the certificate is used in the TLS handshake and how it helps prevent man-in-the-middle attacks.

During the TLS handshake, the certificate is used when the server sends its initial message and follows with the certificate message. The browser then recieves the certificate for www.nhl.com and verifies the issuer, validity dates, and that the domain name is correct. Once the certificate is validated, the browser uses the public key to encrypt and exchange the materials for the session, this how it prevents man-in-the-middle attacks as an attacker can't forge a certificate that's signed by a trusted authority, thus refusing to complete the handshake.

(d) Describe what would happen if the certificate were expired or invalid, including what the browser would show.

If the certificate were expired or invalid, the browser would block the connection. This normally results in a pop-up window stating "Your coinnection is not private" or a network invalid code.

Part 2: VPN Exploration (25 pts)

Explore a VPN explanation site and study how connections are shown.

(a) Describe how a user connects to a website using a VPN and how this differs from a normal connection.

When connecting to a website via a VPN, the device establishes a secure and encrypted connection through the VPN's server. All of the users traffic is then routed through the VPN server before reaching the requested website. This differs from sending traffic directly thoguh the local network and ISP, as with a VPN, only that VPN's server IP Address shows and the users local network can't see the web traffic.

(b) Explain what an "encrypted tunnel" is and how data flows through it.

An encrypted tunnel is the secure connection created between a user's device and the VPN server. All data that passes through the tunnel is encrypted as data flows from the user -> into the tunnel -> to the VPN server -> and then out of the VPN server to the destination website. Return traffic is sent back through the tunnel, just reversed.

(c) Explain the roles of encryption, authentication, and integrity in a VPN, including why symmetric encryption and key exchange are both needed.

- Encryption ensures that data that travels through the VPN can't be viewed/read by any attackers.
- Authentication ensures that the user is connecting to a legitimate VPN server and not a false server.
- Integrity checks ensure that packets are not altered in transit by any attackers.

VPNs use symmetric encryption for the actual data because it's fast and efficient. However, symmetric keys have to be exchanged securely, so VPNs utilize a key exchange protocol to stop from sending the secret key directly over a network.

(d) Compare VPN and TLS in how they protect communication and explain why VPNs are especially useful on public networks.

Both VPNs and TLS encrypt communications but protect different areas of communication. TLS protects only the connection between a browser and specific website while a VPN protects all network traffic to/from the device such as applications, background services, and non-HTTPS sites. VPNs are very useful in public as they prevent local attackers or rouge access points from seeing any user traffic. Even if the public Wi-Fi is not secured, the VPN tunnel keeps all data secured and hidden.

Part 3: Wi-Fi Security (25 pts)

Inspect your Wi-Fi connection settings and identify the type of security being used.

(a) Identify your Wi-Fi security type and explain how your device securely connects to the router.

My device uses WPA2-Personal security on the network named *Duck*. When connecting, the device and router perform an authentication process to verify that both sides know the password to the network without sending the password over the network itself.

(b) Explain the encryption used in modern Wi-Fi and the purpose of the 4-way handshake.

Modern Wi-Fi uses AES-CCMP encryption. The 4-way handshake is used after authentication to prove that both the user/device and the router know the shared secret and generate new session keys. This prevents attackers from reusing old keys and ensures that every device on the network generates its own unique keys.

(c) Explain why wireless communication is more vulnerable than wired communication and describe possible attacks.

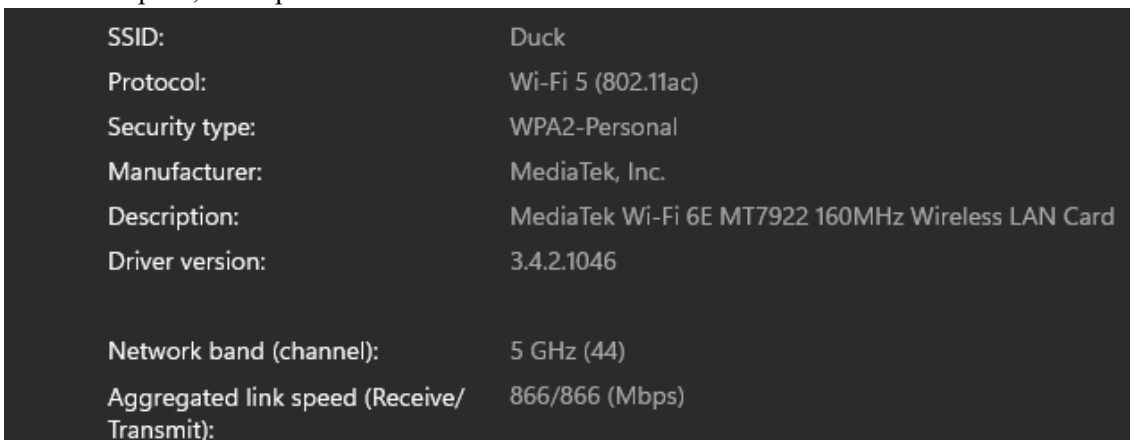
Wireless communication is more vulnerable because signals travel through the air and there is a possibility that an attacker can intercept any communications within range. Attackers don't require physical access to the cables and just need to be close enough.

Possible Attacks:

Eavesdropping
Deauthentication Attacks
Rouge Access Points
Handshake Capture

(d) Explain how WPA3 improves security compared to earlier versions.

WPA3 imrpoves security by replacing WPA2's PSK system with SAE, which protects against offline dictionary attacks. Even if an attacker performs a handshake capture, they can't brute-force the password while offline. WPA3 also enforces stronger encryption, forward secrecy, and protects users even when poor, weak passwords are chosen.



SSID:	Duck
Protocol:	Wi-Fi 5 (802.11ac)
Security type:	WPA2-Personal
Manufacturer:	MediaTek, Inc.
Description:	MediaTek Wi-Fi 6E MT7922 160MHz Wireless LAN Card
Driver version:	3.4.2.1046
Network band (channel):	5 GHz (44)
Aggregated link speed (Receive/ Transmit):	866/866 (Mbps)

Part 4: Kerberos (Conceptual + Investigative) (25 pts)

A university uses Kerberos for login. A student logs in once and can access multiple services. An attacker intercepts a ticket.

(a) Explain how Kerberos enables single sign-on and the role of the Ticket Granting Ticket (TGT), including why passwords are not transmitted.

Kerberos enables single sign-on by allowing a user to authenticate one time and then recieves a TGT from the Authentication Server. The TGT is encrypted with the KDC's sercret key, so only the KDC can read it. Since the TGT has a session key and user's identity, the user is able to request multiple servicess without reentering the password. Passwords are not transmitted over the network, rather it's only used localy to decrypt the response from the KDC.

(b) Explain how tickets, session keys, and authenticators work together during authentication.

When acessing a service, the user has to present the TGT to the TGS to obtain a service ticket. That ticket contains a session key which is shared between the client and service. The client also sends an authenticator, which is encrypted with the session key. The ticket is used to prove the user's identity while the session key establishes and ensures secured communication. The authenticator proves the request is new with the embedded timestamp while also proving it's coming from a legitimate user.

(c) Explain why the intercepted ticket cannot be reused or modified by an attacker, including replay protection.

An intercepted ticket can't be reused or modified because it's encrypted with keys that an attacker does not possess. Even if an attacker were able to intercept a service ticket, they can't decrypt it or extract the session key. The authenticator also contains a timestamp which is valid for only a short period of time, which if attempted by an attacker after expiration, the service will reject the request.

(d) Explain why Kerberos uses symmetric cryptography and what would happen if tickets were not encrypted.

Kerberos uses symmetric cryptography because it's fast and efficient. It also allows the KDC to manage all secret keys. Symmetric encryption ensures that only the KDC, client, and service can read the tickets. If the tickets were not encrypted, attackers could view or modify them, steal session keys, impersonate users, or generate fake service requests.